Application of Modified Goodman Contact Element in Numerical Analysis

Chang-jie Xu, Ru Mou, Yuan-qiang Cai
MOE Key Laboratory of soft Soils and Geoenviromental Engineering
Zhejiang University
Hangzhou, China

Xiao-hua Ma
Key Laboratory of Disaster Prevention and Reduction
Zhejiang Institute of Hydraulics & Estuary
Hangzhou, China

***Abstract*-The Goodman element is widely used to simulate the contact of two different materials, however sometimes it causes large error in stress calculation and element embedding. In this paper, a further modified contact element is proposed, the better element has its own thickness, furthermore, when composed, the element normal connecting form is adopted as the changing mode associated with normal displacement: countless springs would be used as connection when normal displacement is less than thickness, or countless rigid connecting rods would be used. Meanwhile, by comparing the calculation results of horizontal displacement of an actual cutoff wall project using the two different methods with the actual measurement values, the new element has smaller error and is proved to be superior to the exiting Goodman element.**

Keywords-Goodman Elements; Modification; Normal Connection; FEM; Horizontal Displacement

1 Introduction

The interface widely exists in the geotechnical project domain. To effectively simulate interface, many people have conducted the research, also proposed lots of interface models. The four-node Goodman element without thickness was posed in 1968 by Goodman R E, Taylor R L, Brekke T L [1], followed by the thin quadrangle element with thickness by Desai, C S, Zaman, M M in 1984 [2]. Then YIN Zong-ze, ZHU Hong, XU Guo-hua put forward mechanism of rigid-plastic deformation by analyzing the shearing deformation characteristics of the interface between soil and concrete in 1994, based on which they also proposed an interface element with thickness [3]. After that, a new kind of model was built by introducing the concept of "shearing disturbed belt" on the basis of a series of simple shearing tests by ZHANG Dong-lin and LU Ting-hao in 1998 [4]. In 1999, SHAO Wei, JIN Feng and WANG Guang-lun created the no-linear thin element and lead into the regular implantation method, which could take into account both normal and tangential no-linearity [5].

Among these interface element models, Goodman element is used the most frequently for its simple structural form, clear physical relationship and convenience [6]. Such as, He M, Li N, Liu Q simulated the problem of soil slope destruction by exploiting Goodman element [7]; HE Jiang-da, XIE Hong-qiang, WANG Qi-zhi investigated the deformational characteristics of stress-stain in cutoff wall when the tangential stiffness factor of Goodman element changed [8]; LU Jing deprived the element stiffness matrix of Goodman element under large displacement and exerted the matrix into soil-reinforced concrete interface simulation of underground structures [9].

In the normal direction, countless springs are connected for Goodman element. And to consider the interface situation in normal direction, coefficient of elasticity is large, which would cause two problems: first, it is well known that FEM calculation itself has errors when doing calculation, so no wonder, deviations of normal displacement in the calculation could generate. But considering the greatness of coefficient of elasticity of normal springs, the tiny deviations in normal displacement would produce huge effect in normal stress, which could even make the whole calculation meaningless; second, compressive deformations would certainly occur under normal compressive stress, no matter how large the coefficient of elasticity is, which would force the bilateral elements mutual embedding. This is totally against the actual situation of interface. Therefore, some revisions about Goodman element are indispensable.

2 Goodman element and its modification

2.1 Goodman element

Goodman element is a four-node and eight-DOF element without thickness, it is assumed that the two interfaces (12 and 34 exactly) are connected by countless tiny springs in normal and tangential directions.

In Goodman element, mutual coupling of these two directions is not considered, which means the normal stress and shearing stress are only related to normal relative displacement and tangential relative displacement respectively. So the relationship of stress-strain is

$$\{\sigma\}=[k_0]\{\omega\} \tag{1}$$

in which, $\{\sigma\}=\begin{Bmatrix}\tau_s\\ \sigma_n\end{Bmatrix}$;$\{\omega\}=\begin{Bmatrix}\omega_s\\ \omega_n\end{Bmatrix}$;$\{k_0\}=\begin{bmatrix}k_s & 0\\ 0 & k_n\end{bmatrix}$, k_s and k_n are tangential and normal coefficients of elasticity

This work was supported by a grant from the Major Programs of the Science Technology Department of Zhejiang province (No.2008C13040-2)
This work was supported by a grant from the Major Programs of the Water Resources Department of Zhejiang province (No.RA0901)

978-1-61284-774-0/11/$26.00 ©2011 IEEE

respectively.

2.2 Goodman element modification.

Goodman element could well reflect the development of tangential strain-stress in actual interface. However, just as what said before, there are two tremendous drawbacks in Goodman element when doing normal simulation of interface: considerable deviations of normal stress when bearing compression and the mutual embedding of 12-34 interfaces.

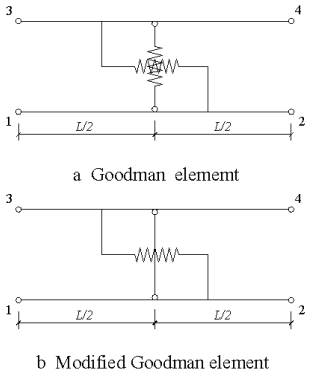

Fig.1. Goodman element and modified model (LI, YU)

To overcome these two drawbacks of Goodman element, LI Shou-de and YU Hong-liang proposed a modified method [10] in which the characteristics of Goodman element without thickness etc are kept and normal springs are changed into rigid connecting rods. When bearing normal compression, no normal relative displacement occurs in the element, and when bearing normal tension, rigid connecting rods break automatically. Through this revision, the drawback that two interfaces would embed each other when bearing compression is conquered.

3 Modification method in this paper

The Goodman element modification method advanced by LI Shou-de and YU Hong-liang could totally overcome the drawback that 12-23 interfaces would embed each other when bearing compression, but it would cause the normal compressive stress more than what it should be. This is because the stiffness of rigid connecting rods is adopted as infinity, to be specific, the two structures connected by these interface elements would be equivalent to confronting the tough with toughness for the lack of buffering of original normal springs in Goodman element, which would certainly increase the original stress.

Meanwhile, when simulating the actual interfaces, both the Goodman element and the modification method by LI and YU could not demonstrate one characteristic that the actual interfaces are with thickness, such as the mudcake between cutoff wall and dam body in earth and rockfill dam, the mudcake between bored pile and surrounding soil. For the existence of mudcake, there would be a cushion between these 12-34 interfaces. So when interfaces bearing compression, the mudcake would deformed by it, which would enlarge the normal relative displacement in these two 12-34 interfaces, but to some extent, reduce the normal stress for buffering function in deformation. Consequently, for the calculation results generated by Goodman element and modification method by LI Shou-de and YU Hong-liang about interface element, the normal displacement would decrease, while the normal stress would increase.

To well simulate the actual foundation, thickness of interface is an important factor which must be considered. Base upon the previous two methods, a new method with further revision is proposed:

1. It is assumed that the thickness between two interfaces is s.

2. The element normal connecting form is adopted as the changing mode associated with normal displacement: countless springs would be used as connection when normal displacement is less than s, or countless rigid connecting rods would be used.

3. Both normal and tangential connections would break when bearing normal tension, so the element stiffness does not make its contribution.

4. The calculating mode of normal stress does not change with normal connecting forms varying when element bearing compression, it would always be $\sigma_n = k_n \cdot \omega_n$

5. Countless tiny springs would be used as connection in tangential direction.

6. The coupling between normal and tangential directions is not considered.

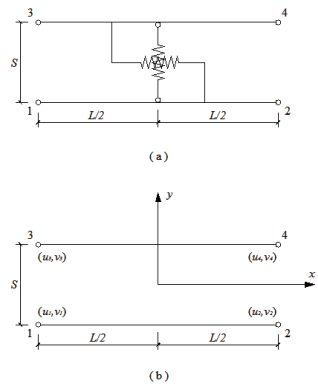

Fig.2 The modified model of Goodman element in this thesis

The specific process is as below:

The tangential displacement for each node is u_1, u_2, u_3, u_4, while the normal displacement is v_1, v_2, v_3, v_4, so the displacement of every node is:

$$u_T = \left(\frac{1}{2}+\frac{x}{L}\right)u_3 + \left(\frac{1}{2}-\frac{x}{L}\right)u_4;\ v_T = \left(\frac{1}{2}+\frac{x}{L}\right)v_3 + \left(\frac{1}{2}-\frac{x}{L}\right)v_4;$$
$$u_B = \left(\frac{1}{2}-\frac{x}{L}\right)u_1 + \left(\frac{1}{2}+\frac{x}{L}\right)u_2;\ v_B = \left(\frac{1}{2}-\frac{x}{L}\right)v_1 + \left(\frac{1}{2}+\frac{x}{L}\right)v_2 \quad (2)$$

The relative displacement is:

In tangential direction

$$w_s = u_B - u_T = \left(\frac{1}{2}-\frac{x}{L}\right)u_1 + \left(\frac{1}{2}+\frac{x}{L}\right)u_2 - \left(\frac{1}{2}+\frac{x}{L}\right)u_3 - \left(\frac{1}{2}-\frac{x}{L}\right)u_4 \quad (3)$$

In normal direction

$$w_n = v_B - v_T = \left(\frac{1}{2}-\frac{x}{L}\right)v_1 + \left(\frac{1}{2}+\frac{x}{L}\right)v_2 - \left(\frac{1}{2}+\frac{x}{L}\right)v_3 - \left(\frac{1}{2}-\frac{x}{L}\right)v_4 \quad (4)$$

Denoted in matrix form as

$$\{w\} = [B]\{\delta\}^e \quad (5)$$

Among which, $\{\delta\}^e = \{u_1\ v_1\ u_2\ v_2\ u_3\ v_3\ u_4\ v_4\}^T$,

if $a = \frac{1}{2} - \frac{x}{L}, b = \frac{1}{2} + \frac{x}{L}$, then

$$B = \begin{bmatrix} a & 0 & b & 0 & -b & 0 & -a & 0 \\ 0 & a & 0 & b & 0 & -b & 0 & -a \end{bmatrix} \quad (6)$$

The element force and the element stress caused by it is assumed as:

$$\{F\}^e = \{U_1\ V_1\ U_2\ V_2\ U_3\ V_3\ U_4\ V_4\}^T \quad (7)$$

$$\{\sigma\} = \begin{Bmatrix} \tau_s \\ \sigma_n \end{Bmatrix} \quad (8)$$

Besides, the virtual displacement of element node and relative displacement of each node in interfaces are assumed as:

$$\left\{\delta^*\right\}^e = \left\{u_1^*\ v_1^*\ u_2^*\ v_2^*\ u_3^*\ v_3^*\ u_4^*\ v_4^*\right\}^T \quad (9)$$

$$\left\{w^*\right\} = \left\{w_s^*\ w_n^*\right\}^T \quad (10)$$

The virtual work by external force is：

$$\begin{aligned} W &= u_1^* U_1 + v_1^* V_1 + u_2^* U_2 + v_2^* V_2 + u_3^* U_3 + v_3^* V_3 + u_4^* U_4 + v_4^* V_4 \\ &= \left\{\delta^*\right\}^{eT} \{F\}^e \end{aligned} \quad (11)$$

The virtual work by internal force is：

$$\begin{aligned} W &= W_s + W_n = \int_{-\frac{L}{2}}^{\frac{L}{2}} s\tau_s w_s^* dx + \int_{-\frac{L}{2}}^{\frac{L}{2}} s\sigma_n w_n^* dx \\ &= s\int_{-\frac{L}{2}}^{\frac{L}{2}} \left[w_s^*\quad w_n^*\right]\begin{bmatrix}\tau_s \\ \sigma_n\end{bmatrix} dx = s\int_{-\frac{L}{2}}^{\frac{L}{2}} \left\{w^*\right\}^T [\sigma] dx \end{aligned} \quad (12)$$

Then what could be derived base on virtual displacement principle is

$$\left\{\delta^*\right\}^{eT} \{F\}^e = s\int_{-\frac{L}{2}}^{\frac{L}{2}} \left\{w^*\right\}^T [\sigma] dx \quad (13)$$

With $\left\{w^*\right\} = [B]\left\{\delta^*\right\}^e$, so

$$\left\{\delta^*\right\}^{eT} \{F\}^e = s\int_{-\frac{L}{2}}^{\frac{L}{2}} \left\{\delta^*\right\}^{eT} [B]^T [\sigma] dx \quad (14)$$

Because $\left\{\delta^*\right\}^{eT}$ is random, so

$$\{F\}^e = s\int_{-\frac{L}{2}}^{\frac{L}{2}} [B]^T [\sigma] dx \quad (15)$$

If putting $\{\sigma\} = [k_0]\{\omega\} = [k_0][B]\{\delta\}^e$ into, then

$$\{F\}^e = s\int_{-\frac{L}{2}}^{\frac{L}{2}} [B]^T [k_0][B]\{\delta\}^e dx = s\int_{-\frac{L}{2}}^{\frac{L}{2}} [B]^T [k_0][B] dx \{\delta\}^e = \{k\}^e \{\delta\}^e \quad (16)$$

Among which

$$\{k\}^e = s\int_{-\frac{L}{2}}^{\frac{L}{2}} [B]^T [k_0][B] dx \quad (17)$$

If doing the integration, then

$$\{k\}^e = \frac{sL}{6}\begin{bmatrix} 2k_s & 0 & k_s & 0 & -k_s & 0 & -2k_s & 0 \\ 0 & 2k_n & 0 & k_n & 0 & -k_n & 0 & -2k_n \\ k_s & 0 & 2k_s & 0 & -2k_s & 0 & -k_s & 0 \\ 0 & k_n & 0 & 2k_n & 0 & -2k_n & 0 & -k_n \\ -k_s & 0 & -2k_s & 0 & 2k_s & 0 & k_s & 0 \\ 0 & -k_n & 0 & -2k_n & 0 & 2k_n & 0 & k_n \\ -2k_s & 0 & -k_s & 0 & k_s & 0 & 2k_s & 0 \\ 0 & -2k_n & 0 & -k_n & 0 & k_n & 0 & 2k_n \end{bmatrix} \quad (18)$$

4 Variation form of normal displacement and normal stress

As previous discussed, increasing the thickness could get the purpose of revising normal displacement and normal stress effectively, and simulate the actual connection situation better. But the normal connection between 12-34 interfaces could change into spring connection or rigid connection due to the situation that whether the thickness of interface element is larger than the normal deformation, which would cause the problem that if the normal stress should change with the normal connecting form varying. If so, the normal connection would change into rigid connection when the calculating value of normal deformation is equal to or more than the thickness of interface. Then, the normal deformation could not go on developing so the thickness of interface is a fixed value as s, then the normal stress would also be a fixed value as $\sigma_n = k_n \cdot s$, which is obviously inconsistent with the fact that the normal stress is increasing all the time under actual force. Hence, it’s necessary to put forward a new corresponding relationship of normal deformation and normal stress.

The varying form concerning node normal deformation and node normal stress adopted in this thesis is shown as Fig.3 followed by specific details:

During the element calculation, the interface states that whether it is bearing tension or compression is distinguished by the node displacement.

If the node normal displacement is $w_n < 0$, it means the interface would crack when bearing tension. To simulate this, the whole normal restrictions of interface elements related to this node should be ignored, besides, the tangential stiffness would be identified as 0.

If the node normal displacement is $0 \le w_n < s$, it means the element is bearing compression and the two interfaces (12-34) do not contact each other directly. Then, spring connection would be used as normal connection and the element stiffness matrix is adopted as Eq. 18. The relationship of normal stress and normal displacement is $\sigma_n = k_n \cdot w_n$.

If the node normal displacement is $w_n \ge s$, it means the element is bearing compression and the two interfaces (12-34) contact one another directly and are mutual embedded. To avoid the mutual embedding, then rigid connecting rod would be used. In this thesis, the node force would be calculated before the node displacement to solve the problem about normal stress. The specific steps are shown below:

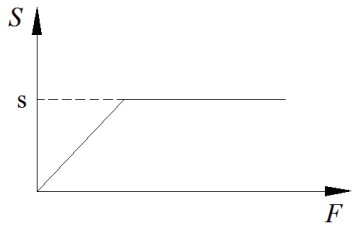

Fig.3 Varying form of normal displacement

1. The node displacement w_n ($w_n \ge s$) is calculated according to the stiffness matrix supplied by Eq. 19;

2. The node stress $\sigma_n = k_n \cdot w_n$ is calculated base on w_n which has been gotten from the first step;

3. The node displacement would be constrained as s if the value turns out more than that.

5 Calculating example

The program of FEM is created base upon the modification method advanced in this thesis [11]. To prove the rationality of this method, the cutoff wall of the earth-rock dam of a certain reservoir in Zhejiang province is calculated [12,13,14,15], furthermore, the generated horizontal displacement values are compared with the measured ones at the same section, and the ones produced by adopting the regular Goodman element.

The dam of this reservoir is the one with thick wall and sand shell. The width of its top and bottom are 10m and 26.6m respectively, the height of the calculated section is 26.6m. And the filling slopes of upstream and downstream are both 1:2 with the filling being gravelly soil. The foundation layers of the dam are sandy loam layer with thickness 5.8m, sand gravel layer with thickness 7.2m, sand shale layer with thickness 1.9m and tuffaceous siltstone layer.

The concrete cutoff wall is located at the place 3.55m from the axis of dam top upstream, with its thickness 80cm, and the bottom of it is embedded into tuffaceous siltstone layer with the depth 0.5m. Besides, the calculated section of this wall is 42.0m high.

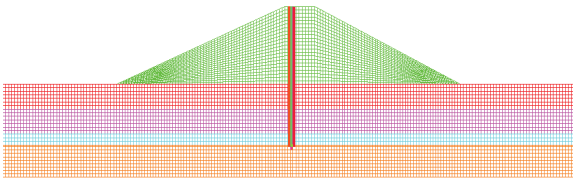
Fig.4 Numerical calculation simulation

Table 1 Parameters of contact element

Contact elements	k_n kPa	s mm
Method in this thesis	4.7×10^5	15
Goodman element	1×10^8	-

The picture of FE mesh is Fig. 4, and the interface element is set between the cutoff wall and the surrounding filling. The horizontal deformation of the wall is computed under the condition that regular operating water level is 20m high.

Parameter values of the interface element could be seen in table 1 for the calculation model, and Duncan-Chang's E-B model is adopted for the soil layers and cutoff wall, whose material parameters could be seen in table 2.

Table 2 Material parameters of E-B model

Materials	c kPa	φ °	K MPa	n	R_f	K_b MPa	m	K_{ur} MPa	n_{ur}
Gravelly soil	23	19	210	0.6	0.85	200	0.5	252	0.6
Sandy loam	20	19	300	0.55	0.8	230	0.4	360	0.55
Sand gravel	0	38	900	0.35	0.7	270	0.2	1080	0.35
Sand shale	3	15	57	0.4	0.75	100	0.6	68	0.4
Tuffaceous siltstone	0	40	2000	0	0	2000	0	2400	0
Cutoff wall	500	30	1050	0	0	1500	0	1260	0

The comparisons between the calculation results generated by the modification method in this thesis and the ones produced by regular

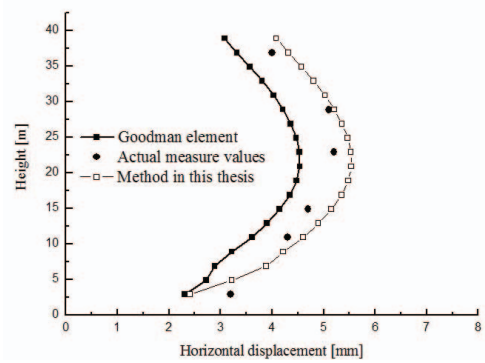

Fig.5 Comparison about wall horizontal displacement

Goodman element, and the measured ones are shown in Fig. 5. In this figure, vertical axis stands for the height of cutoff wall, while horizontal axis is the horizontal displacement of the wall along the axial direction of river bed, with the base point being the bottom of the wall.

The figure could tell us, for horizontal displacement, values generated by the method in this thesis are closer to the measured ones compared with the ones by regular Goodman element. Take the point with measured horizontal displacement in maximum (5.20mm) for example, when it comes to Goodman element, the figure is 4.51mm with the error 13.27%, but when it comes to the method in this thesis, the figure turns out 5.53mm with error 6.34%, which indicates that the method in this thesis is more advantageous and of better application value.

Otherwise, the results of these two methods are only close to one another at the bottom of the wall, for example, when the height is 2.9m, the displacements are 2.3mm and 2.4mm respectively, but at the other heights, the results by the method in this thesis is larger than the ones by Goodman element in value. And the number of this difference would peak at the middle of the wall, for example, the figures are 4.52mm and 5.64mm for the height 21m. As is an explanation, the mudcake between the wall and the filling deforms under compression, which enlarge the displacement of the wall.

6 Conclusions

Goodman element is one with simple structural form, operating convenience and high practicability, but there are two drawbacks which are interface mutual embedding and calculation deviation of normal stress when bearing compression, what's more, these drawbacks may cause the calculation meaningless, which must not be ignored.

In this thesis, Goodman element is modified by increasing the thickness factor, meanwhile, the normal displacement is forced to be the value of element thickness when interface mutually embedding, by which the above drawbacks of Goodman element are overcome efficiently.

Take the cutoff wall of the earth-rock dam of a certain reservoir as investigated object, through comparing the results with the measured ones, and the ones produced by adopting the regular Goodman element, it's revealed that the displacement generated by the method in this thesis is larger than the ones by Goodman element, which suits the deformation characteristic of interface neighboring structures with the thickness increasing. And results of the method in this method fit the measured ones better, which highlights the advantages and application values of this method.

References

[1] Goodman R E, Taylor R L, Brekke T L, "A model for the mechanics of jointed rock," J. Soil Mech. and Found., Engrg.Div, ASCE, vol. 99, pp. 637-660, 1968.

[2] Desai, C S, Zaman M M, "Thin layer element for interface and joints," International Journal for Numerical and Analytical Methods in Geomechanics, vol. 8, pp. 19-43, 1984.

[3] YIN Zong-ze, ZHU Hong, XU Guo-hua, "Numerical simulation of deformation in the interface between soil and structural material," Chinese Journal of Geotechnical Engineering, vol. 16, pp. 15-22, 1994.

[4] ZHANG Dong-ji, LU Ting-hao, "Establishment and application of a interface model between soil and structure," Chinese Journal of Geotechnical Engineering, vol. 20, pp. 62-66, 1998.

[5] SHAO Wei, JIN Feng, WANG Guang-lun, "Nonlinear thin-layer element for modeling interface," J Tsinghua Univ(Sci&Tech), vol. 39, pp. 34-38, 1999.

[6] YIN Zong-ze, ZHU Hong, XU Guo-hua, "A study of deformation in the interface between soil and concrete," Computers and Geotechnics, vol. 17, pp. 75-92, 1995.

[7] HE M, LI N, LIU Q, "Analysis of rock slope stability by using the strength reduction method," Recent Advances and Challenges for the 21st Century. London, Taylor & Francis Ltd, 2008.

[8] HE Jiang-da, XIE Hong-qiang, WANG Qi-zhi, "Sensitivity analysis of tangential stiffness coefficient for contact surface element of different medium," Journal of Sichuan University(Engineering Science Edition), vol. 41, pp. 6-11, 2009.

[9] LU Jing, "Application of Goodman contact element based on large displacement in engineering," West-China Exploration Engineering, vol. 4, pp. 15-19, 2008.

[10] LI Shou-de, YU Hong-liang, "Modification of Goodman interface element," Chinese Journal of Rock Mechanics and Engineering, vol. 23, pp. 2628-2631, 2004.

[11] XIONG Yu-chun, FANG Ying-guang, "Secondary development of material constitutive model in ADINA software," Rock and Soil Mechanics, vol. 29, pp. 2221-2240, 2008.

[12] JIA Li-bin, WANG Zhong-liang, ZHOU Yue-bo, "Theory and research of frictional contact element on concrete diaphragm wall in earth-rock dam," Water Conservancy Science and Technology and Economy, vol. 13, pp. 156-158, 2007.

[13] PENG Cheng-shan, ZHANG Xue-ju, "Effect of friction elements on stress deformation of concrete diaphragm wall of earth-rock dam," Journal of North China Institute of Water Conservancy and Hydroelectric Power, vol. 28, pp. 11-14, 2007.

[14] JIA Li-bin, WANG Zhong-liang, ZHOU Yue-bo, "Theory and research of frictional contact element on concrete diaphragm wall in earth-rock dam," Water Conservancy Science and Technology and Economy, vol. 13, pp. 156-158, 2007.

[15] ZHANG Ga, ZHANG Jian-Min, "Numerical modeling of soil-structure interface of a concrete-faced rockfill dam," Computers and Geotechnics, vol. 36, pp. 762-772, 2009.